Computational Statistics and Data Analysis 168 (2022) 107383

Contents lists available at ScienceDirect

Computational Statistics and Data Analysis

www.elsevier.com/locate/csda

Confidence intervals for parameters in high-dimensional sparse vector autoregression

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A R T I C L E I N F O

Article history:
Received 29 December 2020
Received in revised form 28 September 2021
Accepted 20 October 2021
Available online 25 October 2021

Keywords:
Bootstrap
De-biased Lasso
De-sparsified Lasso
Granger causality
High-dimensional time series

A B S T R A C T

Vector autoregression (VAR) models are widely used to analyze the interrelationship between multiple variables over time. Estimation and inference of the transition matrices of VAR models are crucial for practitioners to make decisions in fields such as economics and finance. However, when the number of variables is larger than the sample size, it remains a challenge to perform inference of the model parameters. The de-biased Lasso and two bootstrap de-biased Lasso methods are proposed to construct confidence intervals for the elements of the transition matrices of high-dimensional VAR models. The proposed methods are asymptotically valid under appropriate sparsity and other regularity conditions. Moreover, feasible and parallelizable algorithms are developed to implement the proposed methods, which save a large amount of computational cost required by the nodewise Lasso and bootstrap. Simulation studies illustrate that the proposed methods perform well in finite-samples. Finally, the proposed methods are applied to analyze the price data of stocks in the S&P 500 index. Some stocks, such as the largest producer of gold in the world, Newmont Corporation, are found to have significant predictive power over most stocks.

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1. Introduction

Vector autoregression (VAR) models have been widely used in macroeconomics, finance, and other fields for their ability to capture the temporal dependence among a set of variables such as stocks in the market (e.g., Sims, 1980; Fuller, 1996; Lütkepohl, 2007). By estimating and inferring the elements of the transition matrices in VAR models, we can evaluate the prediction ability of the interested variables. For instance, Wilms et al. (2016) used the Granger causality test (Granger, 1969), which is equivalent to testing whether each element of the transition matrices is equal to zero or not, to detect the most predictive industry-specific economic sentiment indicators for macroeconomic variables. Lin and Michailidis (2017) proposed testing procedures in multi-block VAR models to test whether a block "Granger-causes" another block of variables and applied them to analyze the temporal dynamics of the S&P 100 component stocks and key macroeconomic variables.

It is common in practice that the number of variables is large, even larger than the sample size. For example, when we study the dynamic relationship between S&P 500 constituent stocks in a year, the number of variables (stocks) is 500, and the number of observations is roughly 252 (trading days). In this high-dimensional setting, the estimation and inference of the parameters of VAR models are challenging. Without additional assumptions, overparameterization and flat likelihood functions hamper the usefulness of the traditional ordinary least square (OLS) estimator (Guo et al., 2016). The more severe

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https://doi.org/10.1016/j.csda.2021.107383
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problem is that the OLS is ill-posed when the number of variables is larger than the number of observations. Inspired by the development of high-dimensional statistics, a series of works proposed sparsity constraints on the transition matrices of high-dimensional VAR models, used the Lasso (Tibshirani, 1996) to estimate the parameters, and established its theoretical properties (Hsu et al., 2008; Song and Bickel, 2011; Chen et al., 2013; Han et al., 2015; Kock and Callot, 2015; Basu and Michailidis, 2015; Davis et al., 2016; Medeiros and Mendes, 2016; Nicholson et al., 2017; Sun et al., 2018; Nicholson et al., 2020; Wong et al., 2020).

Only using a sparse point estimator such as the Lasso does not provide enough uncertainty qualification for decision-making in time series analysis. Taking the example of analyzing the dynamic relationship between S&P 500 constituent stocks, it is desirable to test whether the corrections between the future and current values of the stocks are significant or not. Although the sparsity of the Lasso estimator brings good *estimation* and *prediction* performance (Basu and Michailidis, 2015), its bias and complicated asymptotic distribution make *inference* procedures such as constructing confidence intervals and performing hypothesis testing difficult even for linear regression models with independent observations (Fu and Knight, 2000). Inferences are important for high-dimensional data analysis because they can provide practitioners, such as policy-makers and business owners, with more valuable and solid information for assessing the significance of the correlations between variables or factors. For this purpose, one direction of the research proposed the de-biased Lasso method, which focuses on obtaining an asymptotically normal estimator by correcting the bias of the Lasso estimator (Zhang and Zhang, 2014; van de Geer et al., 2014; Javanmard and Montanari, 2014; Ning and Liu, 2017; Neykov et al., 2018). We will refer the method in Zhang and Zhang (2014) and van de Geer et al. (2014) as LDPE (low dimensional projection estimator) and the method in Javanmard and Montanari (2014) as JM. These estimators are also called de-sparsified Lasso estimators, which sacrifice sparsity to correct the bias and make valid inferences. In particular, when the number of variables is less than or equal to the sample size, the de-biased Lasso estimator is usually reduced to the OLS estimator. Thus, it can be viewed as a generalization of the OLS estimator in high-dimensional settings. Another direction of research used the bootstrap (Efron, 1979) to approximate the asymptotic distribution of the Lasso, such as the bootstrap threshold Lasso (Chatterjee and Lahiri, 2011), bootstrap adaptive Lasso (Chatterjee and Lahiri, 2013), bootstrap Lasso + OLS (Liu and Yu, 2013), and bootstrap de-biased Lasso (Dezeure et al., 2017). Other inference methods include post-selection inference (Berk et al., 2013; Hecq et al., 2019), sample splitting inference (Meinshausen et al., 2009), and so on. We refer to Dezeure et al. (2015) for a comprehensive review on high-dimensional statistical inference methods.

Following the de-biased approach, several researchers proposed methods to constructing confidence intervals and performing hypothesis testing in high-dimensional sparse VAR models (Basu et al., 2019; Zheng and Raskutti, 2019; Babii et al., 2019; Adamek et al., 2020; Deshpande et al., 2020). To improve the finite-sample performance of the de-biased Lasso method, Krampe et al. (2019) proposed a model-based bootstrap de-biased Lasso procedure. However, this method needs to solve hundreds of de-biased Lasso problems, and thus the computational burden is heavy when the number of variables is large. To address this problem, we propose two computationally efficient bootstrap methods, residual bootstrap de-biased Lasso (BtLDPE) and multiplier wild bootstrap de-biased Lasso (MultiBtLDPE). Our contributions are summarized as follows.

First, we derive the asymptotic properties of the de-biased Lasso, residual bootstrap de-biased Lasso and multiplier wild bootstrap de-biased Lasso estimators in high-dimensional sparse VAR models. Additionally, for better finite-sample performance, we modified the degree of freedom in the variance estimator, which is different from that in Adamek et al. (2020) and Deshpande et al. (2020). The most difficult part of the proof is to show that the sparse Riesz condition holds with high probability for high-dimensional sparse VAR models. The dependence structure in the observations of VAR models makes the proof difficult. Moreover, based on the sparse Riesz condition, we prove an upper bound on the sparsity of the Lasso estimator in VAR models, which is novel as far as we know and essential for obtaining appropriate variance estimators and studying the theoretical properties of the bootstrap de-biased Lasso. We highlight these technical difficulties and novelty at the beginning of the proof.

Second, we develop feasible and parallelizable algorithms to implement our methods. Specifically, we perform statistical inference separately for each of the K equations in the VAR models. As the K equations share the same design matrix, we need only to run the nodewise Lasso once, which is the main computational burden of the de-biased Lasso and bootstrap de-biased Lasso. Our methods substantially reduce the computational cost compared to the model-based bootstrap de-biased Lasso method when K is large.

Third, we conduct extensive simulation studies to compare our methods with the bootstrap Lasso, bootstrap Lasso+OLS, and another de-biased Lasso method proposed by Javanmard and Montanari (2014). We find that the LDPE, BtLDPE, and MultiBtLDPE can yield honest coverage probabilities in most cases, and MultiBtLDPE is robust to the heavy-tailed noise and can shorten the interval lengths compared to LDPE. Moreover, we apply our methods to analyze the S&P 500 constituent stocks data set in 2019. We find that the prices of some stocks have significant predictive power over the prices of most stocks.

Notation. For a vector $a = (a_1, \ldots, a_n)$, we denote $\|a\|_1 = \sum_{i=1}^n |a_i|$, $\|a\|_2^2 = \sum_{i=1}^n a_i^2$ and $\|a\|_\infty = \max_{i=1,\ldots,n} |a_i|$. We use e_j to denote a vector whose jth element is one and other elements are zero. For a square matrix A, let $\Lambda_{\min}(A)$ and $\Lambda_{\max}(A)$ be the smallest and largest eigenvalues of A, respectively. Let $|A|$ denote the determinant of A. For a design matrix $\mathbf{X}$, let X_{tj} denote the (t, j)th element of $\mathbf{X}$, X_j denote the jth column of $\mathbf{X}$, and $\mathbf{X}_{-j}$ denote the $\mathbf{X}$ without the jth column. Let $\|\mathbf{X}\|_\infty$ denote the maximum absolute value of the elements of $\mathbf{X}$. For a set S, let $|S|$ denote the cardinality of S and let $\mathbf{X}_S$ denote the columns of $\mathbf{X}$ in S. We use i.i.d as the abbreviation of independent and identically distributed. Let z_α denote the lower α quantile of a standard normal distribution. For two sequences of real numbers $\{a_n\}$ and $\{b_n\}$, we denote $a_n = o(b_n)$ if $a_n/b_n \to 0$ as $n \to \infty$ and $a_n = O(b_n)$ if $\limsup |a_n/b_n| < \infty$. We denote $a_n \asymp b_n$ if there exist positive constants $C_1, C_2 > 0$ such that $\limsup |a_n/b_n| \le C_1$ and $\liminf |a_n/b_n| \ge C_2$. We denote $Z_n = o_p(1)$ if Z_n converges to zero in probability, and $Z_n = O_p(1)$ if Z_n is bounded in probability.

The rest of the paper is organized as follows. In Section 2, we introduce the high-dimensional sparse VAR models and the de-biased Lasso, residual bootstrap de-biased Lasso and multiplier wild bootstrap de-biased Lasso methods. In Section 3, we study the theoretical properties of the proposed methods. In Section 4, we conduct simulation studies to evaluate the finite-sample performance of the proposed methods. In Section 5, we use our methods to analyze the S&P 500 constituent stocks data set. We summarize the results and discuss possible extensions in the last section. All the proofs and additional simulation results are given in the supplementary material.

2. Methods

Consider a K-dimensional VAR(1) model,

$$\mathbf{y}_t = A\mathbf{y}_{t-1} + \mathbf{e}_t, \quad t = 1, \ldots, n, \tag{1}$$

where $\mathbf{y}_t = (y_{1t}, \ldots, y_{Kt})^{\mathrm{T}}$ is a K-dimensional random vector, $A = (a_{ij})_{K\times K}$ is a $K \times K$ transition matrix, and $\mathbf{e}_t = (e_{1t}, \ldots, e_{Kt})^{\mathrm{T}}$ is a K-dimensional white noise whose elements e_{it}'s are independent and identically distributed sub-Gaussian random variables. We have n observations of $\mathbf{y}_t$ and n may be smaller than K. We assume that the VAR process $\{\mathbf{y}_t\}$ is strictly stable.

There are K equations in the VAR(1) model, corresponding to the K components of $\mathbf{y}_t$. We consider each of the K equations separately,

$$y_{it} = a_i^{\mathrm{T}}\mathbf{y}_{t-1} + e_{it}, \quad i = 1, \ldots, K, \quad t = 1, \ldots, n,$$

where $a_i = (a_{i1}, \ldots, a_{iK})$ is the ith row of A. We denote the support set of a_i by $S_i = \{j \in \{1, \ldots, K\} : a_{ij} \neq 0\}$, and let $s_i = |S_i|$. Let $Y_i = (y_{i1}, \ldots, y_{in})^{\mathrm{T}}$, $\mathbf{X} = (\mathbf{y}_0, \ldots, \mathbf{y}_{n-1})^{\mathrm{T}}$, and $\varepsilon_i = (e_{i1}, \ldots, e_{in})^{\mathrm{T}}$, then

$$Y_i = \mathbf{X}a_i + \varepsilon_i, \quad i = 1, \ldots, K. \tag{2}$$

Although equation (2) violates the basic assumptions of linear regression models, it has the same form. Thus, we can apply the de-biased Lasso and bootstrap de-biased Lasso to construct confidence intervals for a_i's, keeping in mind that the validity of these methods need to be further investigated.

2.1. De-biased Lasso

The Lasso is widely used for simultaneous parameter estimation and model selection in high-dimensional linear regression models (Tibshirani, 1996) and VAR models (Hsu et al., 2008; Basu and Michailidis, 2015). It adds an l_1 penalty to the loss function of the OLS estimator to obtain sparse estimates,

$$\hat{a}_i^{\text{Lasso}} := \operatorname*{argmin}_{\alpha \in \mathbb{R}^K}\{\|Y_i - \mathbf{X}\alpha\|_2^2/n + 2\lambda\|\alpha\|_1\}, \quad i = 1, \ldots, K, \tag{3}$$

where λ is a tuning parameter which controls the amount of regularization. In practice, λ can be chosen by cross-validation. We denote the set of selected variables by $\hat{S}_i := \{j \in \{1, \ldots, K\} : \hat{a}_{ij}^{\text{Lasso}} \neq 0\}$ and let $\hat{s}_i := |\hat{S}_i|$. The Lasso estimator is hard to be used for statistical inference because of its bias. Zhang and Zhang (2014) and van de Geer et al. (2014) proposed a method, called the de-biased Lasso, to correct the bias of the Lasso and construct valid confidence intervals for the regression coefficients. This method proceeds as follows.

According to the Karush-Kuhn-Tucker (KKT) condition, the Lasso estimator defined in (3) satisfies

$$-\mathbf{X}^{\mathrm{T}}(Y_i - \mathbf{X}\hat{a}_i^{\text{Lasso}})/n + \lambda\hat{\kappa} = 0, \tag{4}$$

where $\hat{\kappa}$ is the sub-gradient of $\|\alpha\|_1$ evaluated at $\hat{a}_i^{\text{Lasso}}$. It holds that $\|\hat{\kappa}\|_\infty \le 1$ and $\hat{\kappa}_j = \operatorname{sign}(\hat{a}_{ij}^{\text{Lasso}})$ if $\hat{a}_{ij}^{\text{Lasso}} \neq 0$. Let $\hat{\Sigma} := \mathbf{X}^{\mathrm{T}}\mathbf{X}/n$. As $Y_i = \mathbf{X}a_i + \varepsilon_i$, we obtain

$$\hat{\Sigma}(\hat{a}_i^{\text{Lasso}} - a_i) + \lambda\hat{\kappa} = \mathbf{X}^{\mathrm{T}}\varepsilon_i/n. \tag{5}$$

Denote $\Sigma := E(\hat{\Sigma}) = E(\mathbf{y}_{t-1}\mathbf{y}_{t-1}^{\mathrm{T}})$. If we can obtain a good approximation of the precision matrix $\Theta := \Sigma^{-1}$, say $\hat{\Theta}$, then multiplying both hand sides of (5) by $\hat{\Theta}$, we have

$$\hat{a}_i^{\text{Lasso}} - a_i + \hat{\Theta}\lambda\hat{\kappa} = \hat{\Theta}\mathbf{X}^{\mathrm{T}}\varepsilon_i/n + (I - \hat{\Theta}\hat{\Sigma})(\hat{a}_i^{\text{Lasso}} - a_i). \tag{6}$$

The de-biased Lasso estimator is defined as

$$\hat{a}_i := \hat{a}_i^{\text{Lasso}} + \hat{\Theta}\lambda\hat{\kappa} = \hat{a}_i^{\text{Lasso}} + \hat{\Theta}\mathbf{X}^{\mathrm{T}}(Y_i - \mathbf{X}\hat{a}_i^{\text{Lasso}})/n,$$

where the second equality is because of equation (4). Intuitively, as long as the first term of the right-hand of (6) is asymptotically normal with mean zero and the second term is asymptotically negligible, then with a consistent estimator of the variance of ε_i, we can make asymptotically valid inferences for a_i.

Following Zhang and Zhang (2014) and van de Geer et al. (2014), we get $\hat{\Theta}$ by using the nodewise Lasso (Meinshausen and Bühlmann, 2006). Specifically, for $j = 1, \ldots, K$, we run a Lasso regression of X_j versus $\mathbf{X}_{-j}$,

$$\hat{\gamma}_j := \underset{\gamma \in \mathbb{R}^{K-1}}{\operatorname{argmin}}\{\|X_j - \mathbf{X}_{-j}\gamma\|_2^2/n + 2\lambda_j\|\gamma\|_1\}, \tag{7}$$

where λ_j is a tuning parameter. In fact, $\hat{\gamma}_j = \{\hat{\gamma}_{j,k}; k = 1, \ldots, K, k \neq j\}$ is an estimator of $\gamma_j := \Sigma_{-j,-j}^{-1}\Sigma_{-j,j}$, where $\Sigma_{-j,-j}$ is the covariance of $\mathbf{X}_{-j}$ and $\Sigma_{-j,j}$ is the covariance of $\mathbf{X}_{-j}$ and X_j. We denote the sparsity of γ_j by $q_j := |\left\{k \neq j : \gamma_{j,k} \neq 0\right\}|$. We further denote

$$\hat{C} := \begin{bmatrix} 1 & -\hat{\gamma}_{1,2} & \cdots & -\hat{\gamma}_{1,K} \\ -\hat{\gamma}_{2,1} & 1 & \cdots & -\hat{\gamma}_{2,K} \\ \vdots & \vdots & \ddots & \vdots \\ -\hat{\gamma}_{K,1} & -\hat{\gamma}_{K,2} & \cdots & 1 \end{bmatrix},$$

$$\hat{\tau}_j^2 := \left\|X_j - \mathbf{X}_{-j}\hat{\gamma}_j\right\|_2^2/n + \lambda_j\left\|\hat{\gamma}_j\right\|_1,$$

and

$$\hat{T}^2 := \operatorname{diag}(\hat{\tau}_1^2, \ldots, \hat{\tau}_p^2).$$

Then we get

$$\hat{\Theta} := \hat{T}^{-2}\hat{C}. \tag{8}$$

Javanmard and Montanari (2014) proposed a different approach to obtain $\hat{\Theta}$. We do not adopt their approach because of its inferior finite-sample performance as shown in the simulation study.

According to the discussion in Reid et al. (2016), we estimate the variance of ε_i, σ_i^2, by the residual sum of squares divided by its degree of freedom,

$$\hat{\sigma}_i^2 := \frac{1}{n - \hat{s}_i}\|\hat{\varepsilon}_i\|_2^2,$$

where

$$\hat{\varepsilon}_i = (\hat{\varepsilon}_{i1}, \ldots, \hat{\varepsilon}_{in})^{\mathrm{T}} := Y_i - \mathbf{X}\hat{a}_i^{\text{Lasso}}.$$

In contrast, Zheng and Raskutti (2019) uses $\|\hat{\varepsilon}_i\|_2^2/n$ as the variance estimator. We numerically compare the performance of two variance estimators in the supplementary material. Our variance estimator, which corrects the degree of freedom, performs better.

The procedure of nodewise Lasso requires computing the Lasso solution path for K times, which is the main computation burden of the de-biased Lasso. Fortunately, for different variable $i = 1, \ldots, K$, we need only to compute $\hat{\Theta}$ once because the K equations share the same $\mathbf{X}$. The whole procedure is summarized in Algorithm 1, where each `for` loop could run in parallel.

2.2. Bootstrap de-biased Lasso

In this section, we introduce the residual bootstrap de-biased Lasso and wild multiplier bootstrap de-biased Lasso proposed in Dezeure et al. (2017). The only difference between these two bootstrap methods is the approach used to generate bootstrap residuals. For residual bootstrap, we resample with replacement from the centered residuals $\{\hat{\varepsilon}_{it} - \hat{\varepsilon}_{i\cdot}, t = 1, 2, \ldots n\}$, where $\hat{\varepsilon}_{i\cdot} := \sum_{t=1}^{n}\hat{\varepsilon}_{it}/n$, and obtain the bootstrap residuals $\varepsilon_i^* = (\varepsilon_{i1}^*, \ldots, \varepsilon_{in}^*)^{\mathrm{T}}$. For wild multiplier bootstrap, we generate i.i.d multipliers $W_{i1}, \ldots, W_{in}$, independent of the original data, with $E(W_{it}) = 0$, $E(W_{it}^2) = 1$ and $E(W_{it}^4) < \infty$. For example, $W_{it} \sim \mathcal{N}(0, 1)$. Then we multiply the centered residuals by the multipliers, $\varepsilon_{it}^* = W_{it}(\varepsilon_{it} - \hat{\varepsilon}_{i\cdot})$, and obtain the bootstrap residuals $\varepsilon_i^* = (\varepsilon_{i1}^*, \ldots, \varepsilon_{in}^*)^{\mathrm{T}}$.

Next, the bootstrap sample is generated as follows:

$$Y_i^* = \mathbf{X}\hat{a}_i^{\text{Lasso}} + \varepsilon_i^*.$$

Algorithm 1: De-biased Lasso.

Input Data $\{\mathbf{y}_t\}, t = 0, \ldots, n$; Confidence level $1-\alpha$.
For $j = 1, \ldots, K$
  Compute the nodewise Lasso estimator $\hat{\gamma}_j$ and residuals $\hat{Z}_j = X_j - \mathbf{X}_{-j}\hat{\gamma}_j$;
Compute $\hat{\Theta}$ defined in (8);
For $i = 1, \ldots, K$
  Compute the Lasso estimator $\hat{a}_i^{\text{Lasso}}$ using the data $(Y_i, \mathbf{X})$;
  Compute the de-biased Lasso estimator $\hat{a}_i = \hat{a}_i^{\text{Lasso}} + \hat{\Theta}\mathbf{X}^{\mathrm{T}}(Y_i - \mathbf{X}\hat{a}_i^{\text{Lasso}})/n$, residuals $\hat{\varepsilon}_i = Y_i - \mathbf{X}\hat{a}_i^{\text{Lasso}}$, and variance estimator $\hat{\sigma}_i^2 = \|\hat{\varepsilon}_i\|_2^2/(n-\hat{s}_i)$;
  For $j = 1, \ldots, K$
    Compute $l_{ij} = \hat{a}_{ij} - z_{1-\alpha/2}\hat{\sigma}_i\|\hat{Z}_j\|_2/|\hat{Z}_j^{\mathrm{T}}X_j|$ and $u_{ij} = \hat{a}_{ij} - z_{\alpha/2}\hat{\sigma}_i\|\hat{Z}_j\|_2/|\hat{Z}_j^{\mathrm{T}}X_j|$;
Output $1-\alpha$ confidence interval $[l_{ij}, u_{ij}]$ for a_{ij}, $i, j = 1, \ldots, K$.

Algorithm 2: Bootstrap de-biased Lasso.

Input Data $\{\mathbf{y}_t\}, t = 0, \ldots, n$; Confidence level $1-\alpha$; Bootstrap type (residual bootstrap or wild multiplier bootstrap); Number of bootstrap replications B.
For $j = 1, \ldots, K$
  Compute the nodewise Lasso estimator $\hat{\gamma}_j$ and residuals $\hat{Z}_j = X_j - \mathbf{X}_{-j}\hat{\gamma}_j$;
Compute $\hat{\Theta}$ defined in (8);
For $i = 1, \ldots, K$
  Compute the Lasso estimator $\hat{a}_i^{\text{Lasso}}$ using the data $(Y_i, \mathbf{X})$;
  Compute the de-biased Lasso estimator $\hat{a}_i = \hat{a}_i^{\text{Lasso}} + \hat{\Theta}\mathbf{X}^{\mathrm{T}}(Y_i - \mathbf{X}\hat{a}_i^{\text{Lasso}})/n$, residual $\hat{\varepsilon}_i = Y_i - \mathbf{X}\hat{a}_i^{\text{Lasso}}$, and variance estimator $\hat{\sigma}_i^2 = \|\hat{\varepsilon}_i\|_2^2/(n-\hat{s}_i)$;
  For $b = 1, \ldots, B$
    if *Bootstrap type == residual bootstrap* **then**
      Resample with replacement from the centered residuals $\{\hat{\varepsilon}_{ij} - \hat{\varepsilon}_{i\cdot}\}$ and obtain $\varepsilon_i^* = (\varepsilon_{i1}^*, \ldots, \varepsilon_{in}^*)^{\mathrm{T}}$;
    else if *Bootstrap type == wild multiplier bootstrap* **then**
      Generate i.i.d multipliers $W_{i1}, \ldots, W_{in}$ with $E(W_{ij}) = 0$, $E(W_{ij}^2) = 1$, $E(W_{ij}^4) < \infty$, and obtain $\varepsilon_{ij}^* = W_{ij}(\varepsilon_{ij} - \hat{\varepsilon}_{i\cdot})$;
    Generate bootstrap sample $Y_i^* = \mathbf{X}\hat{a}_i^{\text{Lasso}} + \varepsilon_i^*$;
    Compute the bootstrap Lasso estimator $\hat{a}_i^{\text{Lasso}*}$ using $(Y_i^*, \mathbf{X})$;
    Compute $\hat{a}_i^* = \hat{a}_i^{\text{Lasso}*} + \hat{\Theta}\mathbf{X}^{\mathrm{T}}(Y_i^* - \mathbf{X}\hat{a}_i^{\text{Lasso}*})/n$, $\hat{\varepsilon}_i^* = Y_i^* - \mathbf{X}\hat{a}_i^{\text{Lasso}*}$, and $\hat{\sigma}_i^{*2} = \|\hat{\varepsilon}_i^*\|_2^2/(n-\hat{s}_i^*)$;
    Compute the pivot $T_{ij}^{*(b)} = (\hat{a}_{ij}^* - \hat{a}_{ij}^{\text{Lasso}*})|\hat{Z}_j^{\mathrm{T}}X_j|/(\hat{\sigma}_i^*\|\hat{Z}_j\|_2)$;
  For $j = 1, \ldots, K$
    Let $l_{ij} = \hat{a}_{ij} - q_{1-\alpha/2}\hat{\sigma}_i\|\hat{Z}_j\|_2/|\hat{Z}_j^{\mathrm{T}}X_j|$ and $u_{ij} = \hat{a}_{ij} - q_{\alpha/2}\hat{\sigma}_i\|\hat{Z}_j\|_2/|\hat{Z}_j^{\mathrm{T}}X_j|$, where q_α denotes the lower α quantile of $\{T_{ij}^{*(1)}, \ldots, T_{ij}^{*(B)}\}$;
Output $1-\alpha$ confidence interval $[l_{ij}, u_{ij}]$ for a_{ij}, $i, j = 1, \ldots, K$.

Then we replace the original sample $(Y_i, \mathbf{X})$ by the bootstrap sample $(Y_i^*, \mathbf{X})$ to compute the corresponding quantities of the de-biased Lasso. For example, the bootstrap version Lasso estimator is defined as

$$\hat{a}_i^{\text{Lasso}*} := \underset{\alpha \in \mathbb{R}^K}{\operatorname{argmin}}\{\|Y_i^* - \mathbf{X}\alpha\|_2^2/n + 2\lambda^*\|\alpha\|_1\},$$

where λ^* is a tuning parameter. The bootstrap version de-biased Lasso estimator is defined as

$$\hat{a}_i^* := \hat{a}_i^{\text{Lasso}*} + \hat{\Theta}\mathbf{X}^{\mathrm{T}}(Y_i^* - \mathbf{X}\hat{a}_i^{\text{Lasso}*})/n.$$

Note that, one can use different tuning parameters in the original and bootstrap Lasso estimators. Our simulation results indicate that using the same tuning parameters often performs well. We denote the set of variables selected by the bootstrap Lasso as $\hat{S}_i^* := \{j \in \{1, \ldots, K\} : \hat{a}_{ij}^{\text{Lasso}*} \neq 0\}$, and let $\hat{s}_i^* := |\hat{S}_i^*|$. For both bootstrap procedures, we have $E^*(\varepsilon_{it}^*) = 0$ and $\sigma_i^{*2} := E^*(\varepsilon_{it}^{*2}) = \sum_{t=1}^n (\hat{\varepsilon}_{it} - \hat{\varepsilon}_{i\cdot})^2/n$, where E^* denotes the conditional expectation given the original data. Similar to the original Lasso, we estimate σ_i^{*2} by $\hat{\sigma}_i^{*2} := \|\hat{\varepsilon}_i^*\|_2^2/(n-\hat{s}_i^*)$, where $\hat{\varepsilon}_i^* := Y_i^* - \mathbf{X}\hat{a}_i^{\text{Lasso}*}$. We repeat the above procedure B times to obtain the empirical distribution of the statistics of interest.

Because $\mathbf{X}$ is the same in all bootstrap samples, quantities determined by $\mathbf{X}$, such as $\hat{\Theta}$, do not need to be re-computed in the bootstrap replications. This is the most significant advantage of our bootstrap methods over the model-based bootstrap method (Krampe et al., 2019), which needs to re-generate the entire time series $\{\mathbf{y}_t\}$ and re-compute $\hat{\Theta}$ for B times. Moreover, we will show that our proposed bootstrap de-biased Lasso methods are consistent even though they ignore the dependence structure of VAR models. The whole procedure is summarized in Algorithm 2, where each `for` loop could also run in parallel.

3. Theoretical results

In this section, we discuss the theoretical properties of the de-biased Lasso and bootstrap de-biased Lasso. Unlike linear regression models with a fixed design matrix, the design matrix of VAR models, defined in equation (2), is random, exhibits complex dependence structure, and is correlated with the errors $\{\varepsilon_{it}, i = 1, \ldots, K, t = 1, \ldots, n\}$. That is, equation (2) does not justify the assumptions of linear regression models. Fortunately, we can still obtain their asymptotic normality under appropriate conditions, using the deviation bound of the Lasso estimator in high-dimensional sparse VAR models (Zheng and Raskutti, 2019) and martingale central limit theorem (Theorem 5.3.4 in Fuller (1996)).

3.1. Asymptotic distribution of the de-biased Lasso

Our theoretical results require the following assumptions.

Assumption 1. All eigenvalues of A have modulus less than C, where $C < 1$ is a constant that does not change with n and K.

Assumption 2. Suppose that the tuning parameters for the Lasso and nodewise Lasso satisfy: $\lambda \asymp \sqrt{\log(K)/n}$ and $\lambda_j \asymp \sqrt{\log(K)/n}$.

Assumption 3. $\max_i s_i \log(K)/\sqrt{n} = o(1)$.

Assumption 4. $\max_j q_j \log(K)/\sqrt{n} = o(1)$.

Assumption 1 ensures that the time series $\{\mathbf{y}_t\}$ is stationary. Assumption 2 requires that the convergence rates of the tuning parameters are of order $\sqrt{\log(K)/n}$. Assumption 3 is a sparsity assumption on each row of the transition matrix A, which is commonly used in statistical inference based on the de-biased Lasso methods. These two assumptions are the same as those proposed in van de Geer et al. (2014). The sparsity assumption on the precision matrix, Assumption 4, is a little stronger than that in van de Geer et al. (2014). We need this assumption because of the complicated dependence structure in VAR models. Zheng and Raskutti (2019) gave an example where the sparsity of the transition matrix implies the sparsity of the precision matrix; see the following Lemma 1.

Lemma 1. *Suppose that A is symmetric, then we have*

$$q_j \leq \left(\max_i s_i\right)^2 .$$

Proposition 1. *Under Assumptions 1 and 3, for $i = 1, \ldots, K$, there exist constants $0 < c_* < c^* < \infty$, such that, for $s_R > (2 + 2c^*/c_*)s_i + 1$ and $s_R = O(s_i)$, the following holds with probability tending to one:*

$$c_* \leq \min_{\|v\|_0 \leq s_R} \min_{\|v\|_2 = 1} \|\mathbf{X}v\|_2^2/n \leq \max_{\|v\|_0 \leq s_R} \max_{\|v\|_2 = 1} \|\mathbf{X}v\|_2^2/n \leq c^*.$$

Proposition 1 implies that $\mathbf{X}$ satisfies the sparse Riesz condition (Zhang and Huang, 2008), which bounds the restricted eigenvalues of $\mathbf{X}^{\mathrm{T}}\mathbf{X}/n$. Zheng and Raskutti (2019) has obtained the lower bound (see Lemma A.1 in the supplementary material). We complement their result by providing an upper bound. The sparse Riesz condition is crucial for proving our main results.

Theorem 1. *For $i = 1, \ldots, K$,*

(a) Under Assumptions 1–3, we have

$$\|\hat{a}_i^{\text{Lasso}} - a_i\|_1 = O_p\left(s_i\sqrt{\log(K)/n}\right),$$

and

$$\|\mathbf{X}(\hat{a}_i^{\text{Lasso}} - a_i)\|_2^2/n = O_p\left(s_i \log(K)/n\right).$$

(b) Under Assumptions 1 and 3, for any $\lambda \geq 4C_1c^*/c_*\sqrt{\log(K)/n}$, where C_1 is a constant (defined in the supplementary material), we have

$$\hat{s}_i = O_p(s_i).$$

The first statement of Theorem 1 provides the estimation and prediction error bounds of the Lasso estimator, which has been established by Zheng and Raskutti (2019). The second statement provides an upper bound on the sparsity of the Lasso estimator. This bound has been obtained under high-dimensional sparse linear regression models (Zhang and Huang, 2008). Theorem 1 (b) extends the result to high-dimensional sparse VAR models. It is essential for proving the consistency of the variance estimator in the following Theorem 2 and for obtaining the validity of the bootstrap de-biased Lasso. Proposition 4.1 of Basu and Michailidis (2015) provided a similar bound, but for a thresholded variant of the Lasso.

Proposition 2. *Under Assumptions 1, 2 and 4, for $j = 1, \ldots, K$, we have*

$$\|\hat{\gamma}_j - \gamma_j\|_1 = O_p\left(q_j\sqrt{\log(K)/n}\right),$$

and

$$\|\mathbf{X}_{-j}\left(\hat{\gamma}_j - \gamma_j\right)\|_2^2/n = O_p\left(q_j \log(K)/n\right).$$

Proposition 2 provides the estimation and prediction error bounds for the nodewise Lasso estimator.

Theorem 2. *Under Assumptions 1–4, the de-biased Lasso estimator is asymptotically normal; that is,*

$$(\hat{a}_{ij} - a_{ij})/s.e._{ij} \xrightarrow{d} \mathcal{N}(0,1), \quad i, j = 1, \ldots, K,$$

where

$$s.e._{ij} = \frac{\sigma_i \|\hat{Z}_j\|_2}{|\hat{Z}_j^{\mathrm{T}} X_j|}.$$

Furthermore,

$$\hat{\sigma}_i/\sigma_i \xrightarrow{P} 1, \quad i = 1, \ldots, K.$$

Remark 1. The asymptotic normality of the de-biased Lasso has obtained by Zheng and Raskutti (2019) (see Theorem 3.4); however, they used a different variance estimator $\sum_{i=1}^{K} \|\hat{\varepsilon}_i\|_2^2/(nK)$. The conclusion on the variance estimator is new.

Theorem 2 shows that the de-biased Lasso estimator is asymptotically normal and its asymptotic variance can be estimated consistently. Thus, we can construct an asymptotically valid confidence interval for each element of A by using a normal approximation.

3.2. Asymptotic distribution of the bootstrap de-biased Lasso

Interestingly, the bootstrap Lasso estimator has similar convergence rates as the original Lasso estimator. The bootstrap de-biased Lasso estimator is asymptotically normal under the following assumptions.

Assumption 5. Suppose that the tuning parameters for the Lasso, nodewise Lasso, and bootstrap Lasso satisfy: $\lambda \asymp \sqrt{\log(K)/n}$, $\lambda_j \asymp \sqrt{\log(K)/n}$, and $\lambda^* \asymp \log(K)/\sqrt{n}$.

Assumption 6. $s_i(\log(K))^{3/2}/\sqrt{n} = o(1), i = 1, \ldots, K$.

Assumptions 5 and 6 require stronger convergence rates compared to those proposed in Dezeure et al. (2017). However, Dezeure et al. (2017) assumed that $\|\mathbf{X}\|_\infty = O(1)$, which does not hold for the random design matrix $\mathbf{X}$ in VAR models. In fact, we can only show that $\|\mathbf{X}\|_\infty = O_p(\sqrt{\log(Kn)})$. This is why we require stronger conditions on the sparsity and tuning parameters of the bootstrap Lasso estimator.

Theorem 3. *For $i = 1, \ldots, K$,*

(a) Under Assumptions 1, 5 and 6, we have

$$\|\hat{a}_i^{\text{Lasso}*} - \hat{a}_i^{\text{Lasso}}\|_1 = O_p\left(s_i \log(K)/\sqrt{n}\right),$$

and

$$\|\mathbf{X}(\hat{a}_i^{\text{Lasso}*} - \hat{a}_i^{\text{Lasso}})\|_2^2/n = O_p\left(s_i \log^2(K)/n\right).$$

(b) Under Assumptions 1 and 6, for any $\lambda^* \geq 4C_2c^*/c_*\sqrt{\log(K)/n}$, where C_2, c^* and c_* are constants (defined in the supplementary material), we have

$$\hat{s}_i^* = O_p(s_i),$$

where $\hat{s}_i^* = |\{j \in \{1, \ldots, K\} : \hat{a}_{ij}^{\text{Lasso}*} \neq 0\}|$.

Theorem 3 is the bootstrap analogue of Theorem 1. Because we assume different sparsity conditions, the convergence rates in statement (a) of Theorem 3 are different from those in Theorem 1. Now, we can state the main results regarding the validity of the bootstrap de-biased Lasso. Let P^* denote conditional probability given $\{\mathbf{y}_t\}, t = 0, \ldots, n$, and let $\xrightarrow{d^*}$ denote convergence in distribution conditional on $\{\mathbf{y}_t\}, t = 0, \ldots, n$.

Theorem 4. *Under Assumptions 1, 4–6, the bootstrap de-biased Lasso estimator is asymptotically normal; that is, in probability,*

$$(\hat{a}_{ij}^* - \hat{a}_{ij}^{\text{Lasso}})/s.e._{ij}^* \xrightarrow{d^*} \mathcal{N}(0, 1), \quad i, j = 1, \ldots, K,$$

where $\hat{a}_{ij}^*$ denotes either residual or multiplier wild bootstrap de-biased Lasso estimator, and

$$s.e._{ij}^* = \frac{\sigma_i^* \|\hat{Z}_j\|_2}{|\hat{Z}_j^{\mathrm{T}} X_j|}.$$

Furthermore, in probability,

$$\hat{\sigma}_i^*/\sigma_i^* \xrightarrow{P^*} 1, \quad i = 1, \ldots, K.$$

Theorems 2 and 4 imply that the conditional distributions of both residual and multiplier wild bootstrap de-biased Lasso estimators are valid approximations to the (unconditional) distribution of the de-biased Lasso estimator. Thus, we can make valid inferences on a_{ij}'s using the bootstrap.

4. Simulation studies

In this section, we conduct simulation studies to evaluate the finite-sample performance of the proposed methods. We compare our methods with the bootstrap Lasso/Lasso+OLS (Liu and Yu, 2013) and another de-biased Lasso method, JM (Javanmard and Montanari, 2014), in terms of bias and root mean squared error (RMSE) of the estimator, coverage probability and average length of the confidence interval. We also compare our methods with the model-based bootstrap de-biased Lasso method (Krampe et al., 2019) in terms of computational time.

With sample size $n = 100, 150$, dimension $K = 100$ and sparsity $s_i = 3$ for $i = 1, \ldots, K$, we first generate a transition matrix A as follows:

1. We generate $s_i \times K$ non-zero parameters independently from a uniform distribution on $[-1, -0.5] \cup [0.5, 1]$;
2. On each row of A^{init}, we set the diagonal element to be non-zero and randomly arrange the other $s_i - 1$ non-zero parameters on the other positions;
3. As the largest modulus of the eigenvalues of A^{init} might be greater than one, to make the generated time series stable, we let

$$A = \frac{0.6}{\Lambda_{\max}(A^{\text{init}})} A^{\text{init}}.$$

The third step makes the maximum eigenvalue of A equal to 0.6, which ensures the stationarity of the VAR model. The third step will make the absolute value of each element of A small, no matter how large the non-zero parameters are generated in the first step. In our simulation, the range of the absolute values of the non-zero elements of A is $(0.2, 0.4)$. The smaller absolute values of A makes the de-biased Lasso outperform the bootstrap Lasso/Lasso+OLS because the latter requires the "beta-min" condition (all non-zero parameters are sufficiently large in absolute values, see Section 7.4 in Bühlmann and van de Geer (2011)).

We generate data $\{\mathbf{y}_1, \ldots, \mathbf{y}_n\}$ from VAR model (1) with heavy tailed noise $e_{it} \overset{i.i.d}{\sim} t_1$ (t distribution with one degree of freedom, i.e., Cauchy distribution) and $e_{it} \overset{i.i.d}{\sim} t_2$ (t distribution with two degrees of freedom), respectively. We also generate data from VAR model with Gaussian noise $e_{it} \overset{i.i.d}{\sim} \mathcal{N}(0, 1)$ and sub-Gaussian noise $e_{it} \overset{i.i.d}{\sim} U(-1, 1)$. The corresponding simulation results are shown in the supplementary material. We run 500 simulations to evaluate the repeated sampling performance of all methods. We use a bootstrap method described in Section 8 in Ross (2013) to assess the uncertainty

Table 1
Performance of different methods in the case of t_1 distributed noise and $n = 100$.

Type	Method	Bias	rmse	cp	Length	Time
	BtLasso	2.4(0)	4(0)	47(0)	8(0)	–
	BtLassoOLS	1.7(0)	6(2)	57(0)	15(1)	–
	JM	2(1)	40(11)	70(0)	50(3)	–
Non-zero	LDPE	2.3(1)	44(11)	85(0)	81(4)	–
	BtLDPE	2(1)	44(11)	87(0)	78(4)	–
	MultiBtLDPE	2(1)	44(12)	94(0)	83(2)	–
	ModelBtLDPE	2(1)	45(11)	28(0)	102(6)	–
	BtLasso	0(0)	10(2)	96(0)	4(0)	–
	BtLassoOLS	2(1)	48(12)	96(0)	9(0)	–
	JM	21(9)	463(133)	95(0)	72(5)	–
Zero	LDPE	20(8)	453(114)	92(0)	116(7)	–
	BtLDPE	20(9)	450(118)	93(0)	112(6)	–
	MultiBtLDPE	20(9)	448(124)	97(0)	70(4)	–
	ModelBtLDPE	26(11)	571(140)	74(0)	154(11)	–
	BtLasso	2(0)	11(2)	94(0)	4(0)	146
	BtLassoOLS	3(1)	48(11)	94(0)	10(0)	262
	JM	21(9)	464(129)	94(0)	72(5)	17
Overall	LDPE	21(8)	455(108)	92(0)	115(7)	21
	BtLDPE	20(9)	453(123)	92(0)	111(7)	242
	MultiBtLDPE	20(9)	450(121)	97(0)	71(4)	425
	ModelBtLDPE	26(10)	572(144)	73(0)	152(11)	3476

NOTE: bias indicates the aggregate bias; rmse is the root-mean-squared error; cp is the empirical coverage probability; length is the mean interval length; time indicates the total CPU hours required for 500 simulation. The numbers in brackets are the corresponding standard errors estimated using the bootstrap with 200 replications. The cp, length, and their standard errors are multiplied by 100.

across simulations. In particular, after obtaining the 500 bundles results (such as estimates and confidence intervals), we re-sample from them with replacement and calculate the summary statistic, e.g., the root-mean-squared error, using the new bootstrap sample. By repeatedly re-sampling for 200 times, we obtain 200 values of the bootstrap version of the summary statistic. Then, we can use the standard error of these 200 values (we call it Monte Carlo standard error) to approximate the true standard error of the corresponding summary statistic.

We use the `R` package `hdi` to implement the de-biased Lasso (LDPE), residual bootstrap de-biased Lasso (BtLDPE) and multiplier wild bootstrap de-biased Lasso (MultiBtLDPE), `R` package `HDCI` to implement the bootstrap Lasso (BtLasso) and bootstrap Lasso+OLS (BtLassoOLS), `R` code provided by Javanmard and Montanari (2014) to implement their version of de-biased Lasso (JM), and `R` code provided by Krampe et al. (2019) to implement the model-based bootstrap de-biased Lasso (ModelBtLDPE). The tuning parameter is selected by 10-fold regular cross-validation. We also use time-series cross-validation and Bayesian Information Criterion (BIC) to select tuning parameters. Regular cross-validation performs better in our simulation studies; see comparison results in the supplementary material. The same tuning parameter is used for all bootstrap samples. We set the number of bootstrap replications $B = 100$. We use several Intel Xeon E5-2690 V4 processors (2.6 GHz, 35M Cache, 14 Core, 256G Memory) to run the computations for different cases and simulations in parallel, but do not use parallel computations inside the methods (i.e., do not use parallel computations for bootstrap samples or K equations).

Tables 1 and 2 show the simulation results in the case of $n = 100$. The results for $n = 150$ are similar and included in the supplementary material. First, we compare the aggregate bias, i.e., $\{\sum_{i=1}^{K}\sum_{j=1}^{K}(E\hat{a}_{ij} - a_{ij})^2\}^{1/2}$, and RMSE, i.e., $\{E\sum_{i=1}^{K}\sum_{j=1}^{K}(\hat{a}_{ij} - a_{ij})^2\}^{1/2}$ of the Lasso, Lasso+OLS, JM, and LDPE. For non-zero parameters, the Lasso estimator has finite-sample biases, while the Lasso+OLS estimator reduces the bias by 29%–59% and LDPE reduces the bias by 7%–82%. For zero parameters, the Lasso and Lasso+OLS have near-zero biases because of sparsity. Compared to the LDPE and JM, the Lasso and Lasso+OLS have smaller RMSEs for both zero and non-zero parameters. Overall, the RMSEs of the Lasso and Lasso+OLS are 59%–99% smaller than those of the LDPE and JM. Therefore, for *estimation* purposes, we recommend the Lasso and Lasso+OLS because of their smaller overall RMSEs.

Second, we compare the coverage probabilities and average confidence interval lengths of 95% confidence intervals. When the error terms are generated from a t_2 distribution, the JM, LDPE, BtLDPE, MultiBtLDPE nearly reach the nominal confidence level for zero and non-zero parameters. However, the BtLasso and BtLasso+OLS fail to reach the nominal confidence level for non-zero parameters. When the error terms are generated from a t_1 distribution, which is more heavy-tailed, the MultiBtLDPE achieves better coverage probabilities than the LDPE and BtLDPE, especially for non-zero parameters. Moreover, compared to the LDPE and BtLDPE, the MultiBtLDPE reduces the interval lengths by about 13%–44% and 9%–40%, respectively. Since the ModelBtLDPE uses the Gaussian innovations to generate bootstrap samples, it performs poorly in these cases with heavy-tailed noise.

Table 2
Performance of different methods in the case of t_2 distributed noise and $n = 100$.

Type	Method	Bias	rmse	cp	Length	Time
Non-zero	BtLasso	3(0)	3(0)	38(0)	11(0)	–
	BtLassoOLS	2(0)	3(0)	60(0)	21(0)	–
	JM	1(0)	3(0)	92(0)	38(0)	–
	LDPE	0(0)	3(1)	94(0)	41(0)	–
	BtLDPE	0(0)	3(1)	93(0)	39(0)	–
	MultiBtLDPE	0(0)	3(1)	96(0)	50(0)	–
	ModelBtLDPE	1(0)	3(1)	52(0)	91(1)	–
Zero	BtLasso	0(0)	2(0)	95(0)	3(0)	–
	BtLassoOLS	0(0)	5(0)	95(0)	8(0)	–
	JM	1(0)	19(4)	96(0)	40(0)	–
	LDPE	2(0)	20(4)	94(0)	44(0)	–
	BtLDPE	2(0)	20(4)	93(0)	42(0)	–
	MultiBtLDPE	2(0)	20(4)	95(0)	37(0)	–
	ModelBtLDPE	2(0)	20(4)	72(0)	76(1)	–
Overall	BtLasso	3(0)	4(0)	94(0)	3(0)	95
	BtLassoOLS	2(0)	6(0)	94(0)	8(0)	154
	JM	1(0)	19(4)	96(0)	40(0)	10
	LDPE	2(0)	20(4)	94(0)	43(0)	13
	BtLDPE	2(0)	20(4)	93(0)	42(0)	138
	MultiBtLDPE	2(0)	20(4)	95(0)	38(0)	148
	ModelBtLDPE	2(0)	20(3)	71(0)	76(1)	3435

NOTE: bias indicates the aggregate bias; rmse is the root-mean-squared error; cp is the empirical coverage probability; length is the mean interval length; time indicates the total CPU hours required for 500 simulation. The numbers in brackets are the corresponding standard errors estimated using the bootstrap with 200 replications. The cp, length, and their standard errors are multiplied by 100.

Finally, as we expected, the computational cost of the ModelBtLDPE is dozens of times higher than that of the BtLDPE and MultiBtLDPE. Therefore, we recommend the MultiBtLDPE for its honest coverage probabilities, shorter confidence interval lengths, robustness for heavy-tailed noise, and computational feasibility in practice.

5. Real data analysis

In practice, to gain knowledge about the relationship between different stocks, researchers often use VAR models to analyze the stocks and conduct statistical inferences on the elements of the transition matrix. In this section, we use our methods to analyze the price data of the S&P 500 constituent stocks in 2019. There are 505 stocks in total (five companies have two share classes of stock). There are 252 trading days in 2019, but five stocks have incomplete data for some reason. After deleting these five stocks, we have data of 500 stocks for 252 days.

Since our methods require the stationarity assumption, we use the daily return,

$$r_t = \frac{p_t}{p_{t-1}} - 1, \quad t = 1, \dots, n$$

as $\{\mathbf{y}_t\}$ in model (1), where p_t is the adjusted price. We use the augmented Dickey-Fuller test to check the stationarity of the daily return of each stock (Said and Dickey, 1984). All tests reject the null hypothesis of non-stationarity at the significance level of 0.01, indicating no seasonal effects.

We apply LDPE, BtLDPE, and MultiBtLDPE to this data set and obtain the 95% confidence intervals for the elements of the transition matrix A. We also compare our methods with a sparse correlation estimator (SparseCorr) for $\mathbf{y}_t$ and $\mathbf{y}_{t-1}$. For three LDPE-based methods, when the confidence intervals do not contain 0, we consider the corresponding edge of being significant. For SparseCorr, when the estimate is not 0, we consider the corresponding edge of being significant. The second column of Table 4 shows the number of significant edges of all methods without adjustment of multiple testing. Table 3 shows the stocks with the most significant edges. Three LDPE-based methods all agree that the prices of the Newmont Corporation have prediction power on the prices of most stocks. However, the conclusion of SparseCorr is quite different from the other three methods. Because the turbulent global financial environment in 2019 made gold the best safe-haven asset and the Newmont Corporation is the largest producer of gold globally, we believe that the conclusion of the proposed methods is reasonable. Moreover, compared with SparseCorr, which only provides estimates, the proposed methods also measure the uncertainty of estimates for decision-making in practice.

Since there are $500^2 = 250000$ elements in A, we adjust the above results with multiple testing corrections. We also compare our methods with the correlation test (CorrTest) between $\mathbf{y}_t$ and $\mathbf{y}_{t-1}$ without any consideration of temporal dependence. We use four multiple testing correction methods: Bonferroni, BH (Benjamini and Hochberg, 1995), BY (Benjamini and Yekutieli, 2001), and WY (Westfall and Young, 1993) (WY is used only for LDPE related methods). Generally, the Bonferroni method is very conservative, the BH and BY methods are valid only under positive dependence between p-values, and

Table 3
Top 5 stocks with the most predictive power, i.e., the most significant edges.

Method	y_{t-1}	Number of sig
LDPE	NetApp Inc.	189
	Stryker Corporation	172
	BorgWarner Inc.	163
	Keysight Technologies Inc	163
	Newmont Corporation	152
BtLDPE	NetApp Inc.	200
	Keysight Technologies Inc	183
	Stryker Corporation	183
	Newmont Corporation	173
	BorgWarner Inc.	172
Multi-BtLDPE	Newmont Corporation	185
	BorgWarner Inc.	163
	Stryker Corporation	162
	Johnson & Johnson	153
	Kinder Morgan Inc Class P	136
SparseCorr	Tapestry Inc.	182
	Qorvo Inc.	175
	Motorola Solutions Inc.	162
	Micron Technology Inc.	141
	Advanced Micro Devices Inc.	137

Table 4
Number of significant edges and run time of different methods.

Method	Unadj	Bonferroni	BH	BY	WY	Time
LDPE	14780	187	283	105	189	0.735
BtLDPE	19619	0	0	0	55	3.041
MultiBtLDPE	16882	0	0	0	1	2.817
SparseCorr/CorrTest	12513	169	516	123	–	0.002

NOTE: BH stands for Benjamini and Hochberg (1995); BY stands for Benjamini and Yekutieli (2001); WY stands for Westfall and Young (1993). Time indicates the total CPU hours required for corresponding method.

Table 5
False discovery rates, recall rates, and F scores of different methods.

Method	Avg FDR (%)	Avg recall rate (%)	Avg $F_{0.05}$ score (%)
LDPE	77.0	1.3	12.1
BtLDPE	53.2	1.0	13.7
MultiBtLDPE	12.8	0.5	8.8
CorrTest	79.3	1.5	12.3

the WY method is recommended by Dezeure et al. (2017) under the i.i.d. setting. Table 4 shows the number of significant edges produced by different methods. Among them, BtLDPE + WY has a moderate number of significant edges. The detailed information of significant edges of BtLDPE + WY is given in the supplementary material.

Finally, to evaluate the validity of the considered methods, we generate a simulated data set based on the real data to compare the performance of different methods. Specifically, we use the Lasso estimator of A as the ground truth, whose sparsity $s_i \approx 5$. The largest modulus of the eigenvalues of A is smaller than one. We use the empirical distribution of residuals as the noise, which is close to t_5 distribution (see Fig. 1). By repeatedly sampling with replacement from $\hat{\varepsilon}_{it}$, we generate 50 replications according to the VAR model (1). Then, we examine the repeated sampling properties of the proposed methods on this semi-real data set. We use the false positive rate (FDR), recall rate (RR), and F score as evaluation metrics. They are defined as

$$\mathrm{FDR} = \frac{n_{\mathrm{fp}}}{n_{\mathrm{tp}} + n_{\mathrm{fp}}}, \quad \mathrm{RR} = \frac{n_{\mathrm{tp}}}{n_{\mathrm{tp}} + n_{\mathrm{fn}}}, \quad F_\beta = (1+\beta^2)\frac{(1-\mathrm{FDR}) \times \mathrm{RR}}{\beta^2(1-\mathrm{FDR}) + \mathrm{RR}},$$

where n_{fp}, n_{tp}, and n_{fn} are the numbers of false positive, true positive, and false negative, respectively. F score combines the performance of FDR and RR. The smaller β is, the more weight is assigned to FDR. Table 5 shows that the LDPE and correlation test approach have high FDR, and the BtLDPE performs best in terms of F score.

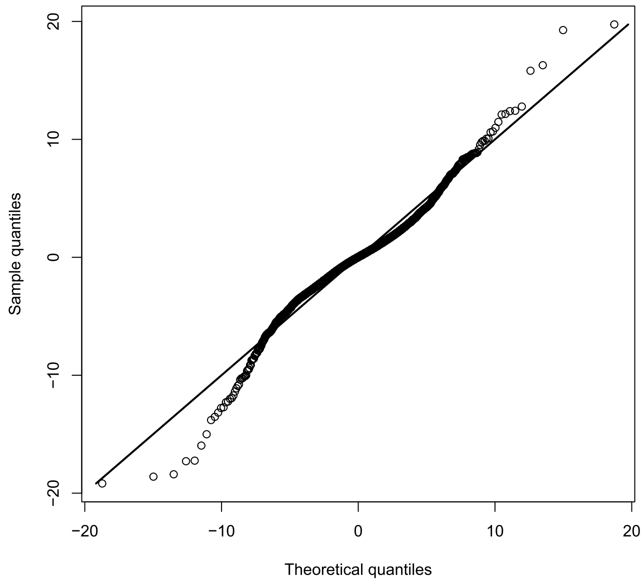

Fig. 1. Q-Q plot of empirical quantile of the residuals and theoretical quantile of t_5 distribution.

6. Conclusion

Performing statistical inference for parameters in high-dimensional VAR models is a challenging but essential problem. In this paper, we propose the de-biased Lasso (LDPE), residual bootstrap de-biased Lasso (BtLDPE), and multiplier wild bootstrap de-biased Lasso (MultiBtLDPE) to construct a confidence interval for each element of the transition matrix in VAR models. Unlike linear regression models with a fixed design matrix, the design matrix in VAR models is random with a complex dependence structure, making theoretical analysis of the proposed methods challenging. Based on the convergence rates of the Lasso and nodewise Lasso estimators, we derive the asymptotic unbiasedness of the de-biased Lasso estimator. Combined with the martingale central limit theorem, we obtain its asymptotic normality. We also derive the asymptotic properties (conditional on $\{\mathbf{y}_t\}, t = 0, \ldots, n$) of two bootstrap de-biased Lasso methods and demonstrate their validity for statistical inferences in high-dimensional sparse VAR models. Furthermore, we propose feasible and parallelizable algorithms to implement our methods. More specifically, we apply the de-biased Lasso, residual bootstrap de-biased Lasso and multiplier wild bootstrap de-biased Lasso to each of the K equations in VAR models separately. As the K equations share the same design matrix, we need only to compute the nodewise Lasso once, which is the main computational burden of the de-biased Lasso and bootstrap de-biased Lasso methods. Compared to the model-based bootstrap method, the proposed bootstrap methods substantially reduce the computational cost, especially when K is large.

We conduct comprehensive simulation studies to compare our methods with the bootstrap Lasso, bootstrap Lasso+OLS, and another de-biased Lasso method proposed by Javanmard and Montanari (2014). We find that the LDPE, BtLDPE, and MultiBtLDPE can produce honest coverage probabilities when the noise distribution is not severely heavy-tailed. Moreover, the MultiBtLDPE is robust to heavy-tailed distributed noise and shortens the confidence interval lengths compared to the LDPE and BtLDPE. Therefore, we recommend the MultiBtLDPE for statistical inferences.

This paper focuses on statistical inferences for each element of the transition matrix in high-dimensional sparse VAR models. It would be interesting to extend the methods for constructing simultaneous confidence intervals and carrying out multiple hypothesis testing. Our numerical results show that Westfall-Young procedure (Westfall and Young, 1993) might be helpful. We leave the corresponding theoretical investigation to future work.

Acknowledgements

The authors thank the editor, associate editor, and three reviewers for helpful comments. The authors thank Dr. Lixiang Zhang for his suggestions that have helped clarify the text. Dr. Hanzhong Liu's research is supported by the National Natural Science Foundation of China (grant no. 11701316).

Appendix A. Supplementary material

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.csda.2021.107383.

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